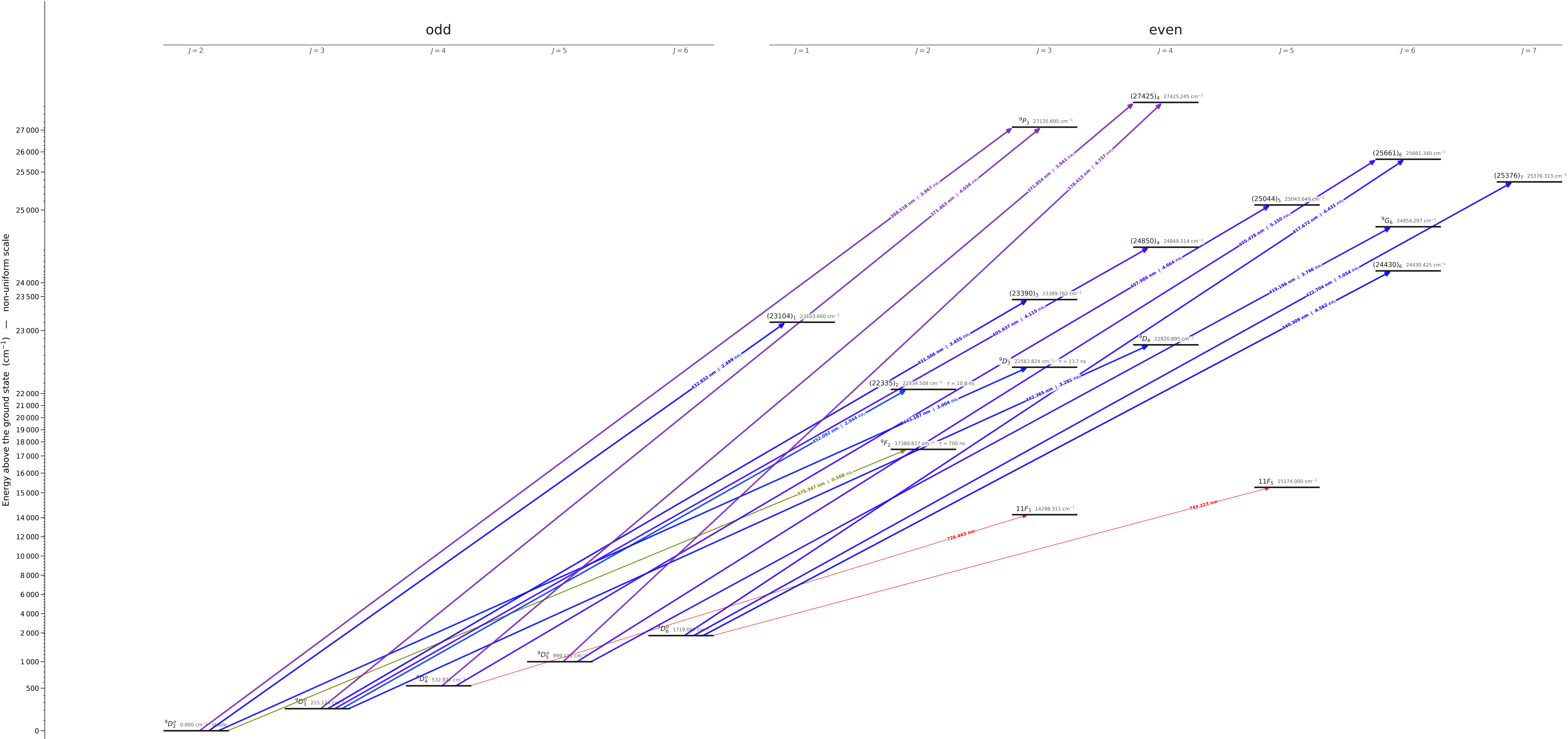
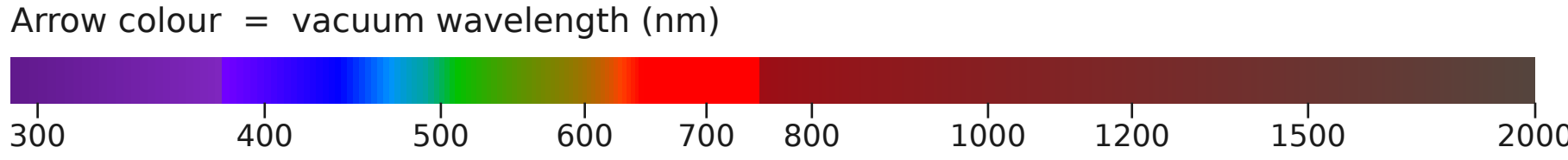


Gd (Gd I). Strongest classified transitions below 2 μm (NIST): 21 levels, 19 lines.

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	(THz)	$ \langle J e J' \rangle $ (ea0)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$9D_2^o - 9P_3$	368.5183	368.4134	813.5077	3.967	9.100e+07	-	-	proposed laser-cooling line
$9D_3^o - 9P_3$	371.4631	371.3575	807.0584	4.036	9.200e+07	-	-	371 nm repumper for the proposed 368 nm cooling scheme
$9D_6^o - (25376)_7$	422.7038	422.5848	709.2258	7.054	8.900e+07	-	-	proposed laser-cooling line
$9D_2^o - 9D_3$	443.1873	443.0629	676.4464	3.004	3.000e+07	11.62	41.1	443.06 nm
$9D_3^o - (22335)_2$	452.0922	451.9654	663.1224	2.944	3.800e+07	14.74	41	451.96 nm
$9D_2^o - 9F_2$	575.3466	575.1871	521.0641	0.568	6.855e+05	0.2274	48	first step
$9D_4^o - 11F_3$	726.4626	726.2624	412.6743	-	-	-	-	726.266 nm spin-forbidden line $9D^4 - 11F_3$ studied as a c...
$9D_6^o - 11F_5$	743.2229	743.0183	403.3681	-	-	-	-	743.014 nm spin-forbidden line $9D^6 - 11F_5$ studied as a c...

 $1/2\pi\Gamma$: natural linewidth of the upper level, $1/(2\pi\Gamma)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
$0D_0^2$	47.950	191.160	F=1/2: -48.51	F=3/2: -143.85	F=5/2: -143.45	F=7/2: +191.64
$11F_3$	-77(5)	590(4)	F=1/2: +813.30	F=5/2: +179.68	F=7/2: -294.75	F=9/2: -197.32
$11F_5$	84(4)	967(4)	F=7/2: -335.94	F=9/2: -587.13	F=11/2: -303.02	F=13/2: +871.08
$9F_2$	95.100	347.000	F=1/2: -124.33	F=3/2: -285.30	F=5/2: -264.43	F=7/2: +372.05

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$0^0_2 - ^0_2$	575.347	152-156	2232(0)
$0^0_2 - ^0_2$	575.347	154-156	1116(0)
$0^0_2 - ^0_2$	575.347	155-156	763(0)
$0^0_2 - ^0_2$	575.347	157-156	-21(0)
$0^0_2 - ^0_2$	575.347	158-156	-1119(0)
$0^0_2 - ^0_2$	575.347	160-156	-2233(0)
$0^0_4 - 11\text{f}_3$	726.463	158-160	964(4)
$0^0_4 - 11\text{f}_3$	726.463	157-160	2154(20)
$0^0_4 - 11\text{f}_3$	726.463	156-160	2563(4)
$0^0_4 - 11\text{f}_3$	726.463	155-160	3243(12)
$0^0_6 - 11\text{f}_5$	743.223	158-160	1054(4)
$0^0_6 - 11\text{f}_5$	743.223	157-160	486(65)
$0^0_6 - 11\text{f}_5$	743.223	156-160	2029(4)
$0^0_6 - 11\text{f}_5$	743.223	155-160	2476(12)